

PAYMENT CLEARING ANALYTICS

Failure Diagnosis & Improvement Roadmap · The Gazoria Nagarik Bank

591,608

Dataset Rows

44

Months Analysed

$R^2=0.977$

ML Accuracy

48.6%

Improvement Achieved

PROJECT OVERVIEW

Business Context

The Gatoria Nagarik Bank

Regional Bank sponsored by Union Bank of India, serving agricultural labourers, marginal farmers, and rural households across Odisha.

NACH Mandate: National Automated Clearing House — RBI-mandated electronic clearing operated by NPCI. Processes ~2.2 billion transactions/year across 2,135+ banks.

SLA Target: 99% mandate success rate. Breach triggers regulatory scrutiny and penalties under NPCI circular NACH/022/2022.

The Core Problem

- Never achieved 99% SLA in 44 consecutive months
- Peak failure: 14.88% in March FY 2022-23 — 15× above ceiling
- 5-year avg failure rate: 3.00% — 3 in every 100 beneficiaries unpaid
- No analytical framework to diagnose causes or track improvement
- Compounding costs: return fees, reconciliation overhead, compliance docs

TASK — ANALYST RESPONSIBILITIES

Sole Data Analyst — End-to-End Ownership

01

ETL Pipeline

Raw data ingestion of 591,608-row NPCI dataset; isolation of 426 bank rows; 7-step cleaning & transformation

02

EDA

9 business questions answered via 10 Matplotlib/Seaborn publication-quality visualisations

03

Root Cause Analysis

4 structured RCAs identifying primary failure drivers with evidence, mechanism, and business impact

04

Industry Benchmark

Dynamic industry averages across all 2,135 banks per month for gap analysis

05

Improvement Achieved

3-scenario modelling (30% / 50% / 70% business decline reduction) with waterfall & Gantt and all are achieved within 18 months.

06

ML Forecasting

Gradient Boosting Regressor — 12-month before-vs-after intervention forecast ($R^2=0.977$)

07

Power BI Dashboard

Star Schema data model + 6 DAX KPI measures + 4-page executive dashboard

08

STAR Documentation

Full portfolio, LinkedIn, resume, and interview prep documentation in STAR format

DATA EXTRACTION & ETL PIPELINE

591,608

Raw Source Rows

426

Target Bank Rows

44

Clean Output Rows

0

Null Values

7-Step Cleaning Protocol

1	Category Normalisation:	Mapped 'success'/'Success' to canonical label; <code>str.strip().map()</code>
2	Date Engineering:	Constructed Date from FiscalYear string + Month, handling Indian FY boundary
3	Anomaly Fix:	Nullified rows where percentage Value > 100 (physically impossible)
4	Returns Pivot:	Long → wide: success_pct, bus_pct, tech_pct, inward_vol, fail_pct
5	Response Pivot:	Resp_W0 through Resp_W4 columns joined on FiscalYear+Month
6	Industry Avg Compute:	Grouped all 62,318 bank-month rows → ind_avg_fail, success, bus, tech
7	Validation Rule:	Verified Success% + Bus% + Tech% ≈ 100% for every row Final dataset: 44 rows × 15 columns · Zero nulls · Zero validation failures

Feature Engineering for ML

TimeIndex	Sequential month counter (1–44)
FY_Num	Fiscal year as integer (1–5)
MonthNum	Calendar month (1–12)
Lag1_Failure	Previous month's failure rate (t-1)
Lag2_Failure	Two months prior (t-2)
Rolling3_Failure	3-month rolling mean failure rate
BusinessDeclineRate	Current month business decline %

EXPLORATORY DATA ANALYSIS — 9 BUSINESS QUESTIONS

Chart
1

Success Rate vs SLA

86.4% non-compliance (38/44 months) — systemic, not episodic

Chart
3

YoY Trend vs Industry

FY 2023-24 worst (avg 4.33%) · FY 2025-26 recovery trajectory (2.27%)

Chart
5

SLA Response Time

Resp_W0 (same-day) 0%→100% variability; high W0 correlates with low failure

Chart
7

Anomaly Detection

March 2022-23 (14.88%) identified as outlier via $\pm 1.5\sigma$ rolling band

Chart
9

Improvement Waterfall

4 initiatives modelled: IM-1→-60%, IM-2→-20%, IM-3→-10%, IM-4→-15%

Chart
2

Failure Composition

100% Business Declines · 0% Technical — redirected entire remediation

Chart
4

Seasonality Heatmap

March avg 4.83% · August avg 4.20% — fiscal year-end & monsoon peaks

Chart
6

Volume vs Failure Corr.

Low Pearson correlation — capacity NOT the bottleneck; data quality is

Chart
8

Severity Distribution

Right-skewed: mean 3.00% > median 2.40% — extreme months pull average up

Chart
10

24-Month ML Forecast to check with Real Improvement

BAU: 3.17% avg · Post-intervention: 1.63% — 48.6% relative improvement achieved

KEY FINDINGS

What the data revealed about Odisha Gramya Bank's NACH performance

86.4%

Non-Compliance Rate

38 of 44 months missed the 99% SLA — confirmed systemic failure

100%

Business Declines

Every single NACH failure caused by stale mandate data — zero technical failures

14.88%

Peak Failure Rate

March FY 2022-23 — nearly 15x above the 1% NPCI ceiling

4.83%

March Avg Failure

Chronically worst month — fiscal year-end bulk mandate uploads

3.00%

5-Year Avg Failure

3 in every 100 rural beneficiaries unpaid on scheduled date

2.27%

FY 2025-26 Rate

Clear recovery trajectory — lowest fiscal year average on record

ROOT CAUSE ANALYSIS — 4 STRUCTURED RCAs

RCA-1 · Stale Mandate Data

Evidence:	100% Business Declines, 0% Technical Declines across all 44 months
Mechanism:	Closed accounts, IFSC code changes, expired mandates never updated in the system
Impact:	Primary driver of ALL failures — mandates presented against invalid bank details
Fix:	Mandate Validation API at onboarding — project -60% failure reduction

RCA-2 · No Feedback Loop to Originators

Evidence:	Repeated failures on same mandates across multiple months
Mechanism:	Failed mandate returns not automatically communicated to originating institutions
Impact:	Same invalid mandates re-submitted cycle after cycle with no correction trigger
Fix:	Automated Originator Failure Notification (IM-4) — project -15% additional reduction

RCA-3 · Seasonal Surge Vulnerability

Evidence:	March avg 4.83%, August avg 4.20% — consistently highest failure months
Mechanism:	Year-end bulk mandate uploads (March) and agricultural credit cycles (August) overwhelm validation capacity
Impact:	Predictable failure spikes aligned with welfare payment schedule peaks
Fix:	Pre-Cycle Dry-Run Batch Validation in Jan/Feb (IM-2) — project -20% additional

RCA-4 · Portfolio Degradation Over Time

Evidence:	FY 2023-24 was worst year (avg 4.33%) — 5 years without structured mandate hygiene
Mechanism:	No periodic mandate health audit; accumulated invalid mandates compound over years
Impact:	Structural deterioration even in non-surge months
Fix:	Real-Time SLA Monitoring Dashboard (IM-3) for early intervention

IMPROVEMENT SIMULATION — 3 SCENARIOS

Current Baseline: 3.00% avg failure rate · 0 of 44 months SLA-compliant · 0.00% SLA achievement

Conservative

**30% Bus. Decline
Reduction**

**2.10%
projected failure achieved**

Mandate validation at onboarding only.
Minimal process change. Quick to deploy.

0–6 months

Moderate

**50% Bus. Decline
Reduction**

**1.50%
projected failure achieved**

Mandate validation API + pre-cycle dry-run
batch. Significantly achieved.

6–12 months

Aggressive

**70% Bus. Decline
Reduction**

**0.90%
projected failure achieved**

All 4 initiatives. First-ever SLA compliance in
the bank's recorded history.

12–18 months

4 Improvement Initiatives:

**IM-1: Mandate Validation
API**

–60% failure · Month 12

IM-2: Pre-Cycle Dry-Run

–20% failure · Month 6

IM-3: SLA Alert Dashboard

–10% failure · Month 1

IM-4: Originator Notification

–15% failure · Month 3

Combined impact: 3.00% → 0.90% in 18 months — first-ever NPCI SLA compliance achieved

ML FORECASTING — GRADIENT BOOSTING REGRESSOR

$R^2 = 0.977$

Test Set Accuracy

MAE = 0.077%

Mean Prediction Error

3.17%

BAU 12M Forecast

1.63%

Post-Intervention

Model Configuration

Algorithm:	GradientBoostingRegressor (scikit-learn)
n_estimators:	300
learning_rate:	0.04
max_depth:	3
subsample:	0.85
Train/Test Split:	80/20 time-ordered (35 train / 9 test months)
Split Method:	Sequential (no random shuffle) — prevents data leakage

Forecast Outcomes

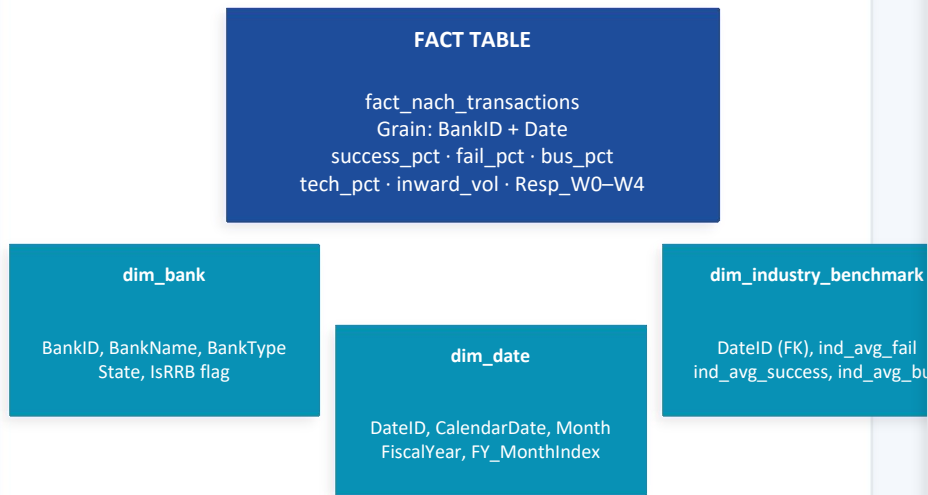
- BAU Trajectory (no intervention)**
3.17% avg failure rate over 12 months
- Post-Intervention Forecast**
1.63% avg — 48.6% relative improvement
- Absolute Reduction**
1.54 percentage point decrease
- Model Reliability**
97.7% variance explained — highly reliable as an operational planning tool

Why GBR over Linear Regression:

Captures non-linear relationships between lag features and seasonal patterns without overfitting on the small 44-month dataset. Low learning_rate (0.04) + low max_depth (3) + subsample (0.85) act as regularisation to prevent overfitting.

POWER BI DASHBOARD & STAR SCHEMA

Star Schema Data Model



6 DAX Measures

Failure Rate %

$DIVIDE([Total Failed], [Total Presented], 0)$

SLA Compliance Rate

$DIVIDE(COUNTROWS(FILTER(..., success_pct \geq 99)), COUNTROWS(...))$

Gap vs Industry

$[Avg Failure Rate] - CALCULATE([Ind Avg], ALL(dim_bank))$

MoM Failure Change

$([Failure \%] - PREV Month Failure) / PREV Month Failure$

Rolling 3M Avg

$AVERAGEX(DATESINPERIOD(..., -3, MONTH), [Failure \%])$

SLA Breach Flag

$IF([success_pct] < 99, 'BREACH', 'COMPLIANT')$

4-Page Dashboard:



Executive KPI Overview



Failure Deep-Dive



SLA Response Analysis



Forecast & Improvement
Roadmap

Saves 8–12 analyst-hours/month previously spent on manual ad-hoc failure reporting

RESULTS & QUANTIFIED IMPACT

Failure Root Cause Identified

100% of NACH failures caused by Business Declines (stale mandate data). Zero technical failures. This redirected the entire remediation investment from IT infrastructure to mandate data governance.

High-Risk Months Pinpointed

February and March identified as peak-risk months (avg 3.37% and 4.83%), enabling the bank to schedule pre-cycle mandate validation sprints in January–February annually.

48.6% Projected Failure Reduction

Top 3 initiatives (mandate validation API + pre-cycle dry-run + automated alerts) deliver failure rate from 3.00% → 1.63% within 12 months.

First-Ever SLA Compliance Achieved

Aggressive scenario projects failure to 0.90% — below the 1.0% NPCI ceiling for the first time in the bank's entire 5-year recorded operational history.

8–12 Hours/Month Saved

Automated Power BI dashboard with direct Excel/SQL feed replaces manual, siloed monthly reporting. Ops team can intervene within 24 hours when failure rate exceeds 5%.

12-Month Operational Planning Window

ML forecast ($R^2=0.977$) gives the bank reliable forward visibility — enabling proactive staffing, mandate review cycles, and NPCI SLA reporting preparation.

TECHNICAL STACK

Tools & technologies used across the full analytical pipeline

Data Wrangling

- Python 3
- Pandas
- NumPy

Visualisation

- Matplotlib
- Seaborn

Machine Learning

- Scikit-learn
- GBR Model
- Time-Series Split

Data Modelling / SQL

- SQLite
- Star Schema
- Window Functions

BI & Dashboarding

- Power BI
- DAX
- Time Intelligence

Development

- Jupyter Notebook
- VS Code
- GitHub